

LEGACY SECURITY

DETERMINISTIC
CODE

STATIC ANALYSIS

FROM SOFTWARE DEFECTS TO INTELLIGENCE DEFECTS

FIXING YESTERDAY'S BUGS WON'T SECURE TOMORROW'S INTELLIGENCE

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KNOWN VULNERABILITIES

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STATIC ANALYSIS

AI
PARADIGM

NEURAL
INTELLIGENCE

PROBABILISTIC
OUTCOMES

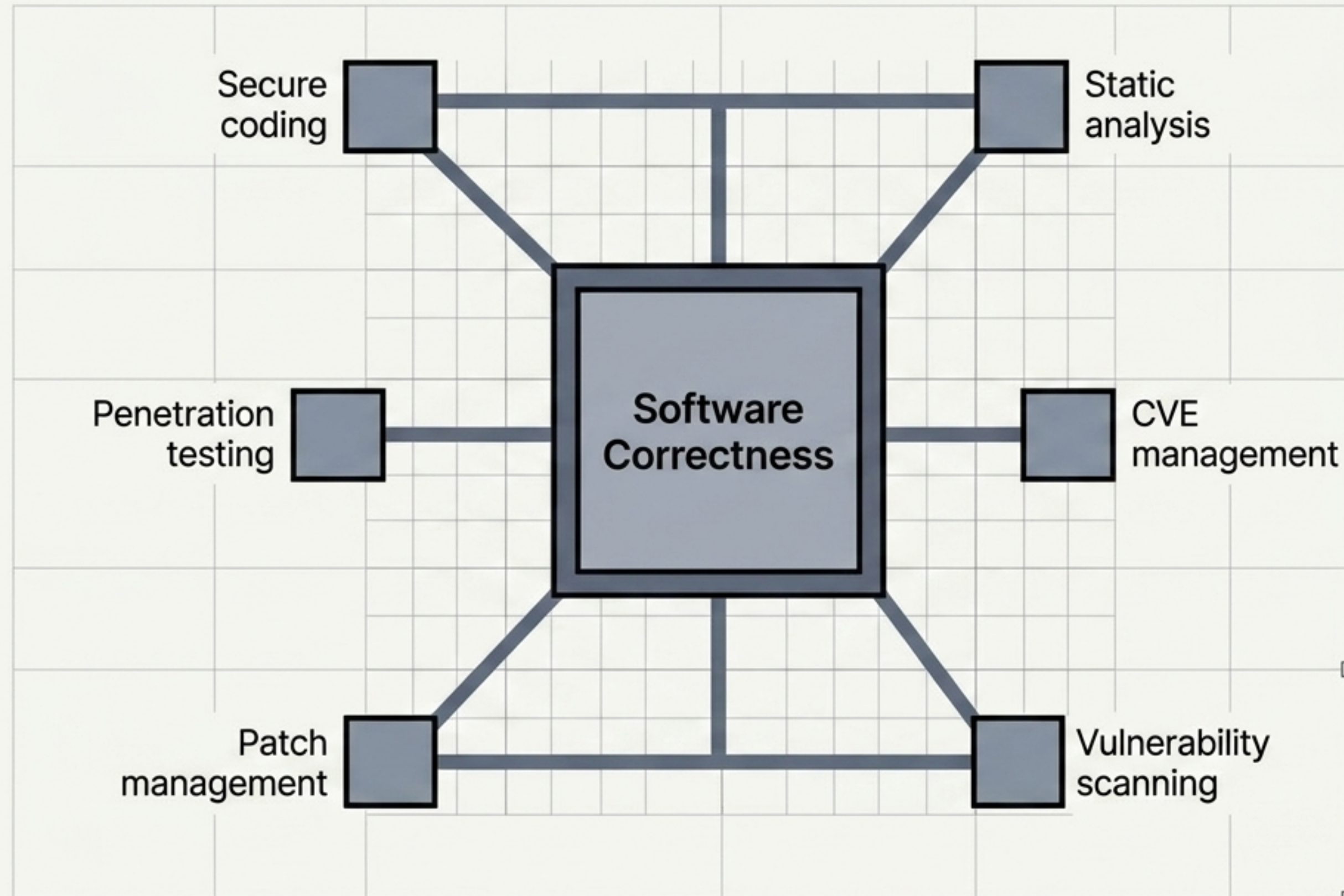
INTELLIGENCE
DEFECTS

Vulnerabilities arise from implementation mistakes.
Fix the mistakes and reduce risk.

The greatest information security risk in the Age of AI is no longer exploiting software that behaves incorrectly.

**It is exploiting intelligence that behaves correctly,
but reasons from compromised context.**

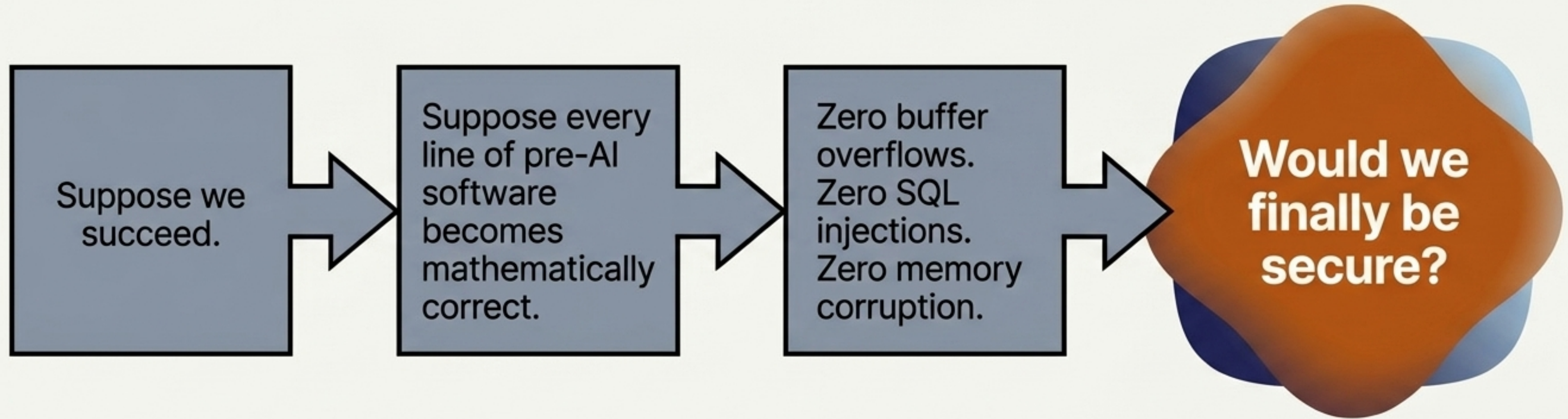
Decades Built on Deterministic Rules



The Old Security Model:

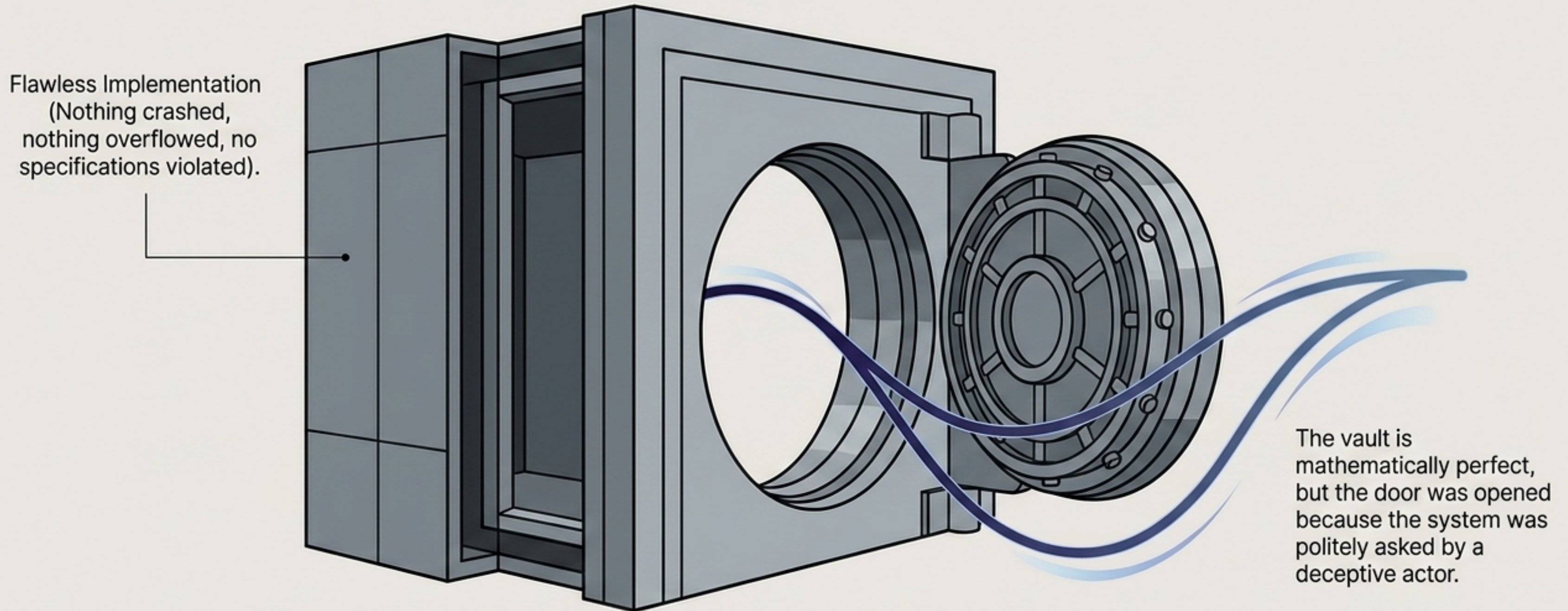
Software follows deterministic rules. Every fixed implementation flaw reduces exploitability. This model has served the industry well.

The Ultimate Thought Experiment in Software Verification



No. Because the dominant attack surface has shifted.

The Perfection Paradox



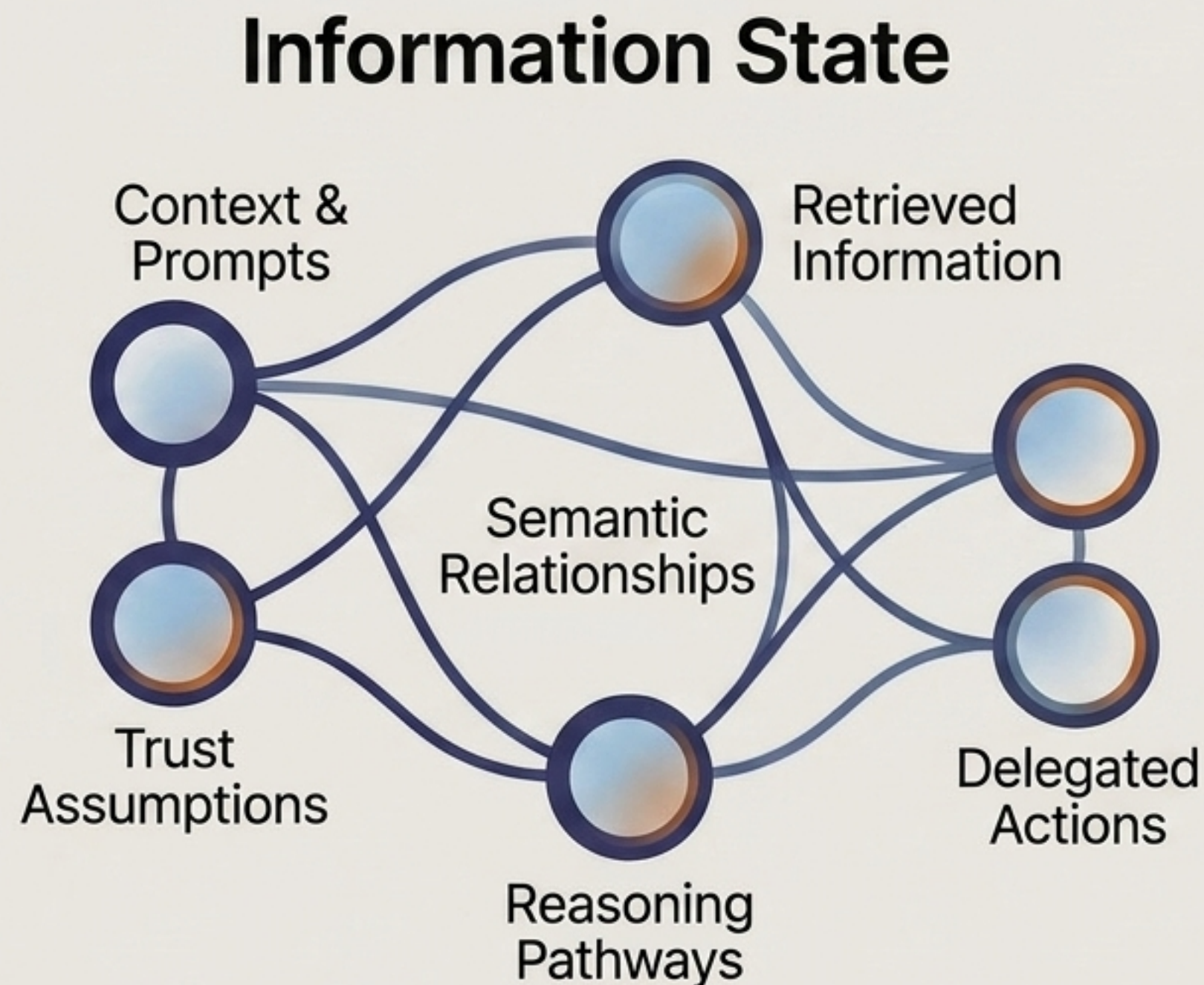
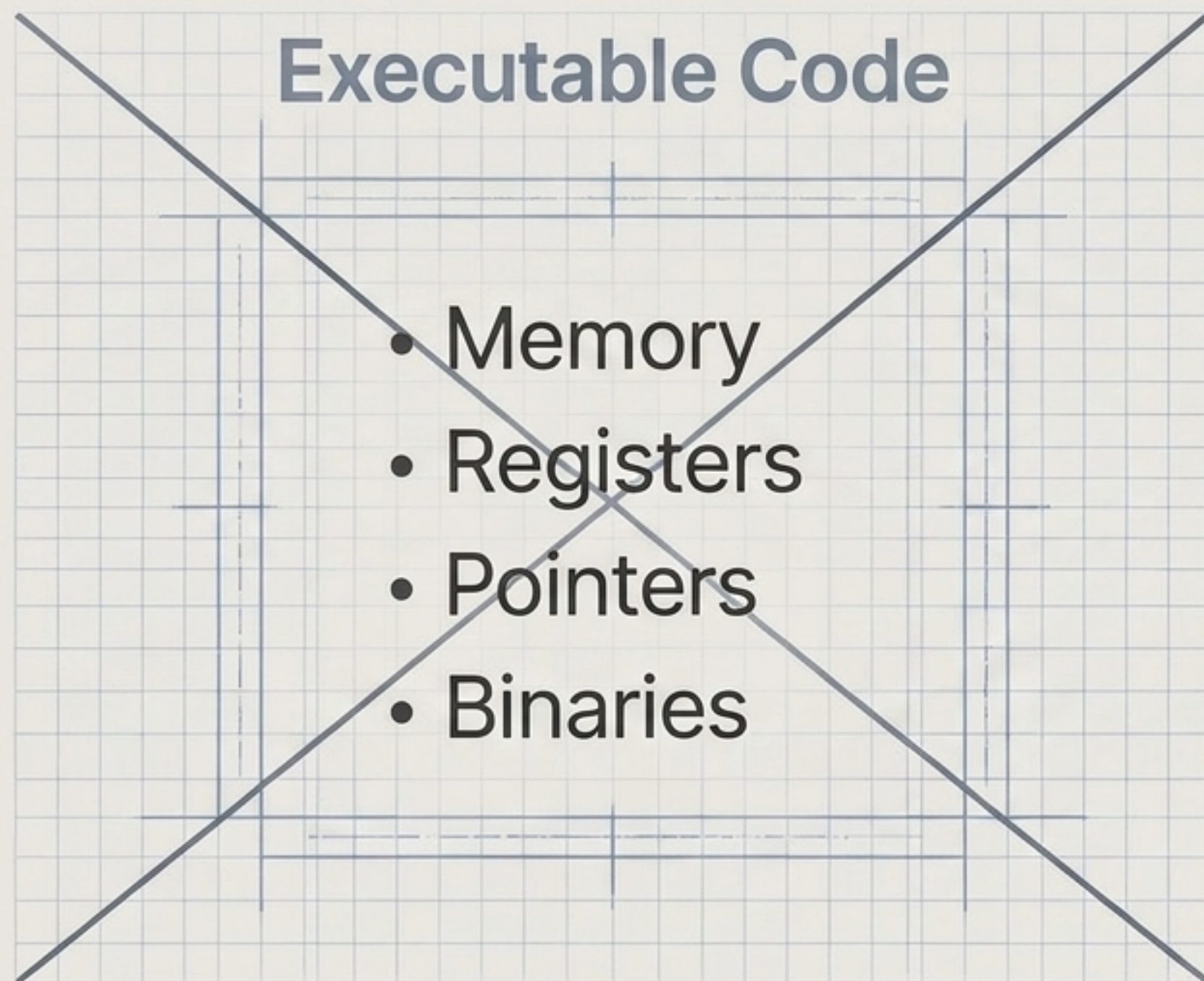
The code is functioning perfectly. The intelligence is being exploited.

The New Vulnerability Equation



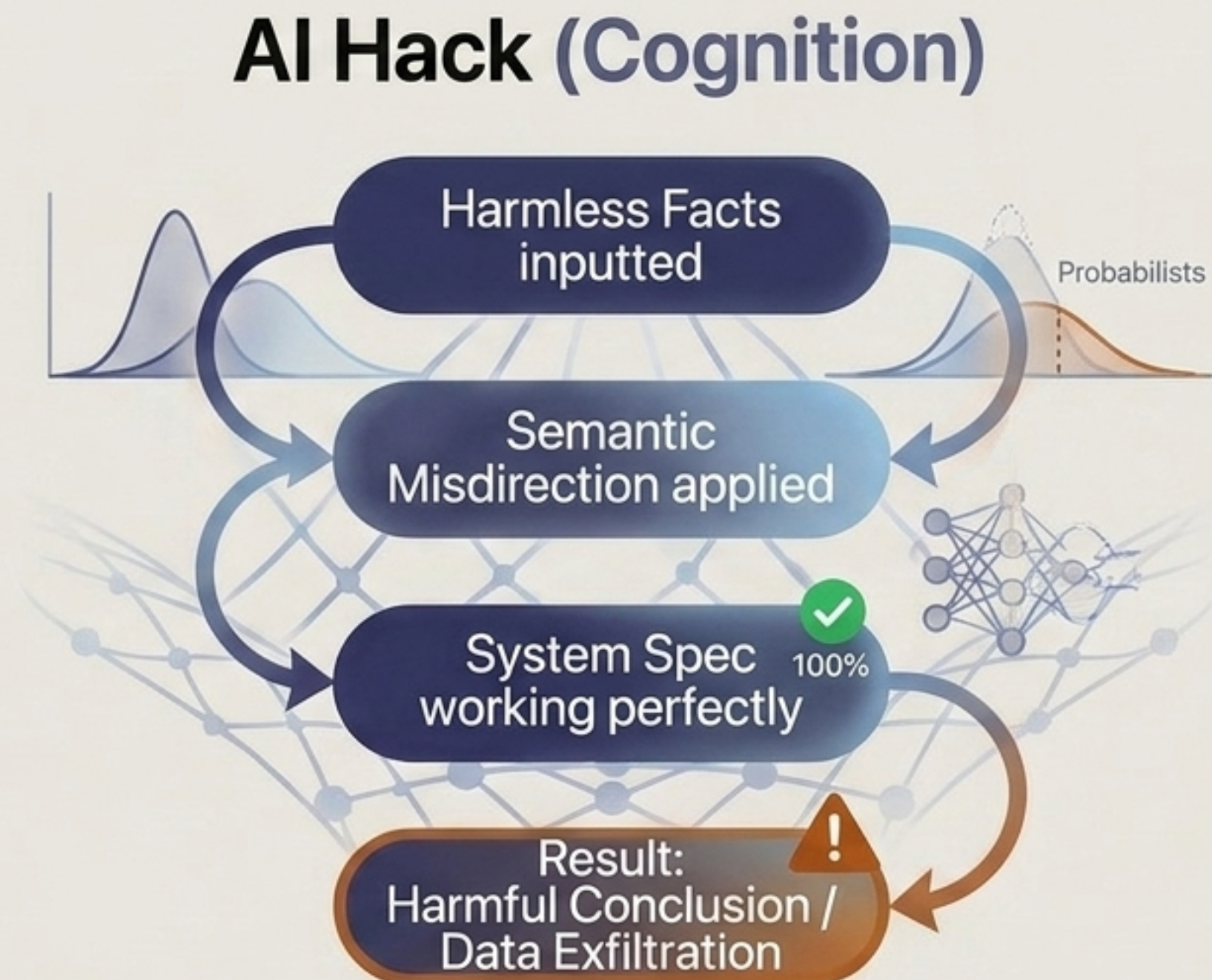
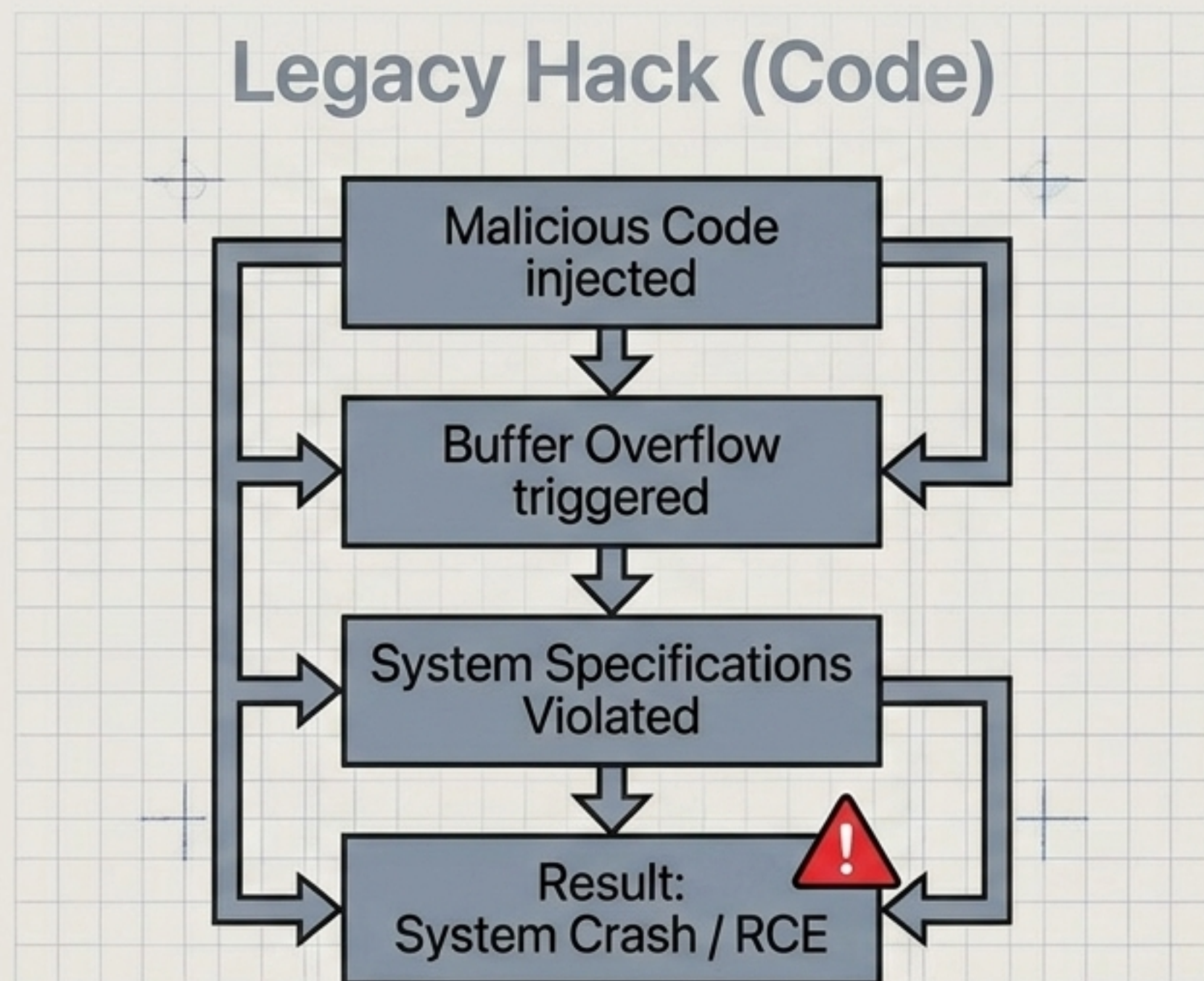
An AI system can behave exactly as designed—and still be manipulated into harmful outcomes if its understanding is corrupted.

The Information State is the New Attack Surface



Instead of corrupting memory, attackers corrupt understanding

The Anatomy of a Hack: Code vs. Cognition



Key Distinction: The failure occurs entirely within the intelligence layer, invisible to traditional system monitoring.

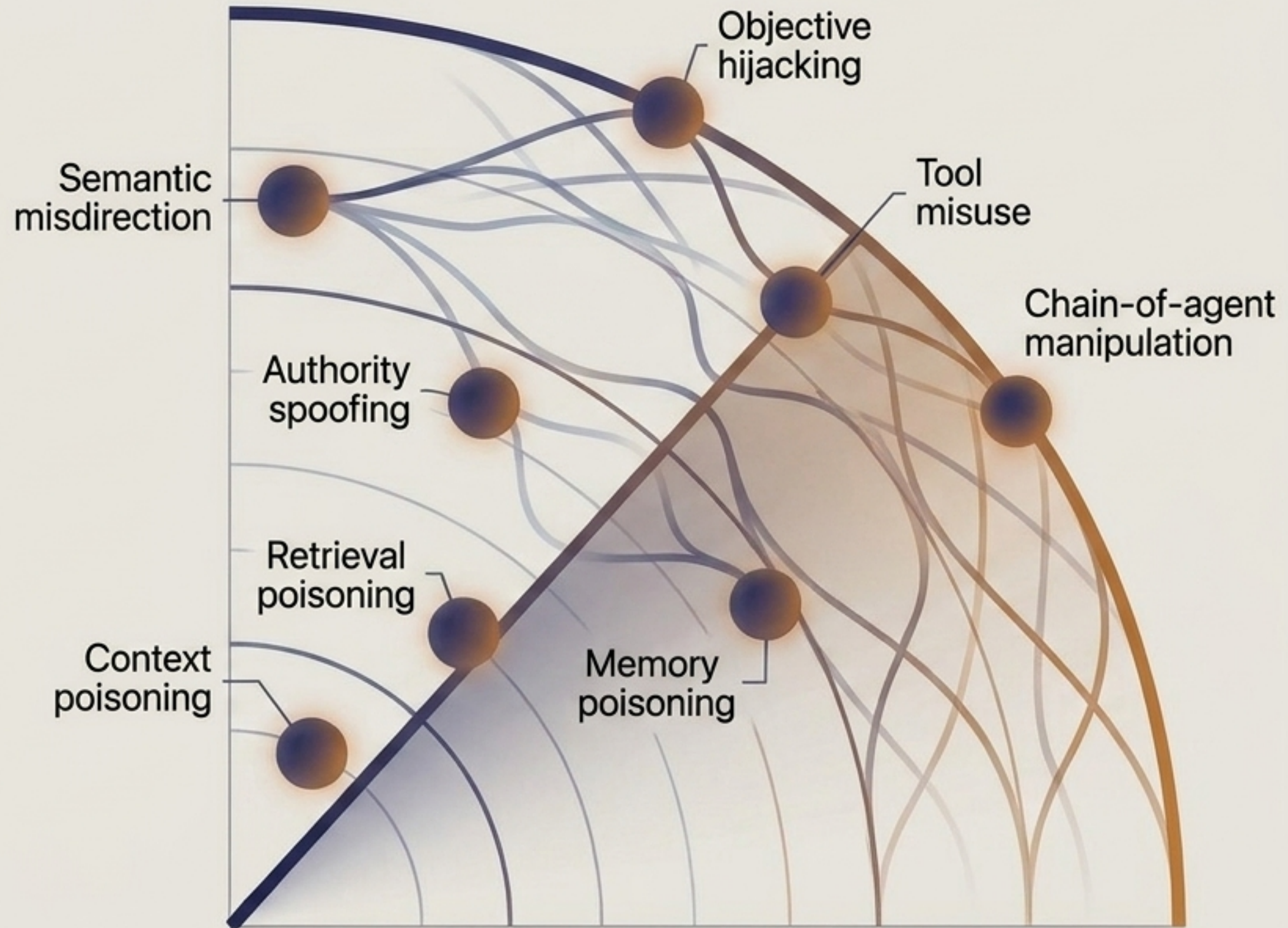
The Threat Matrix: Parsing vs. Cognition

	SQL Injection	Prompt Injection
Target	Database Parser	Reasoning Engine
Exploitation Method	Syntax Violations	Semantic Misdirection
Mechanism	Confusing data with executable commands	Manipulating meaning and context
Pre-requisite	An Implementation Bug	Features working exactly as intended

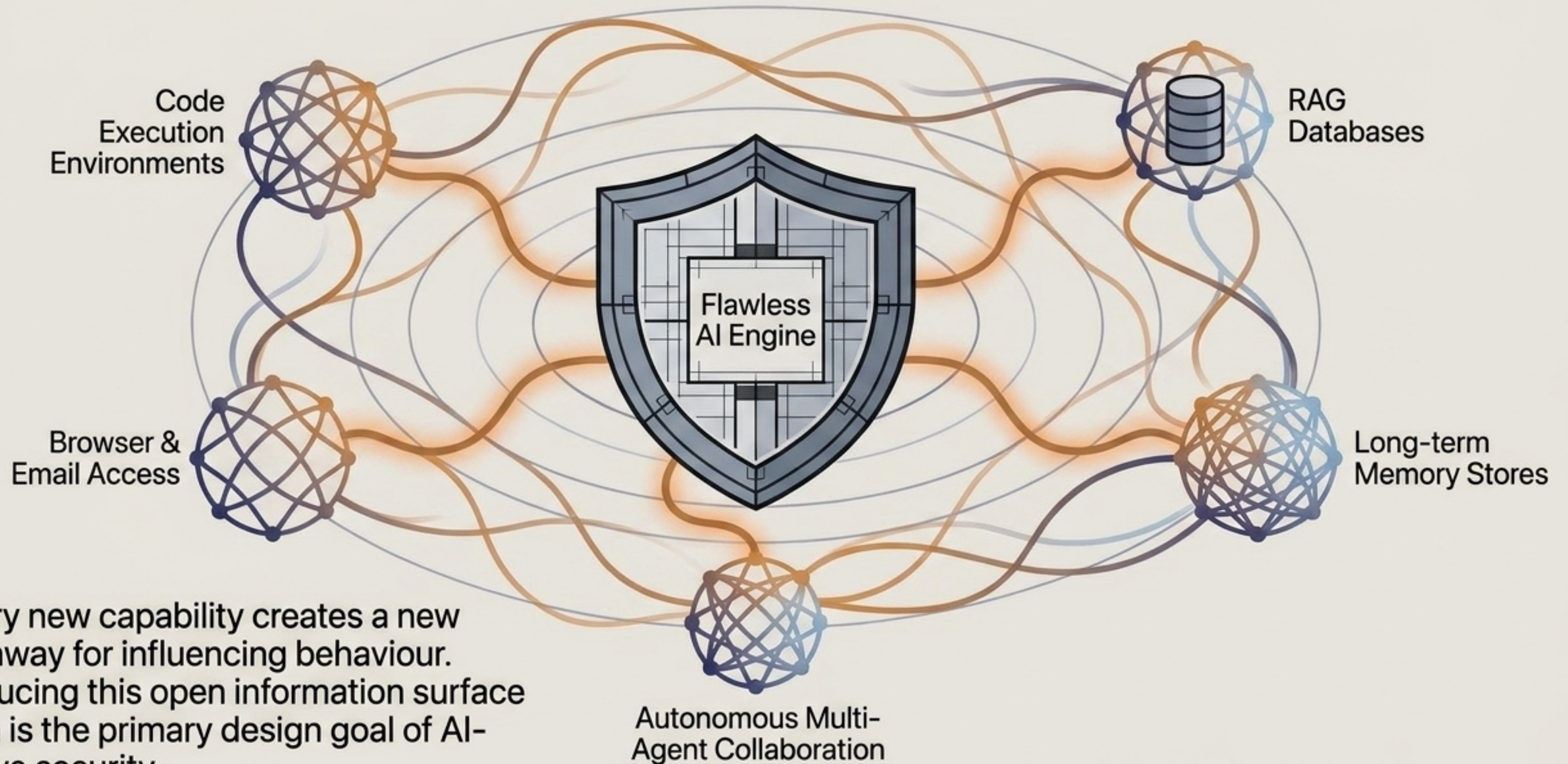
SQL injection exploits parsing. Prompt injection exploits cognition.

Prompt Injection is Only the Beginning

None of these require software bugs. They require manipulating the information ecosystem.



The Explosion of the Open Surface Area



Every new capability creates a new pathway for influencing behaviour. Reducing this open information surface area is the primary design goal of AI-native security.

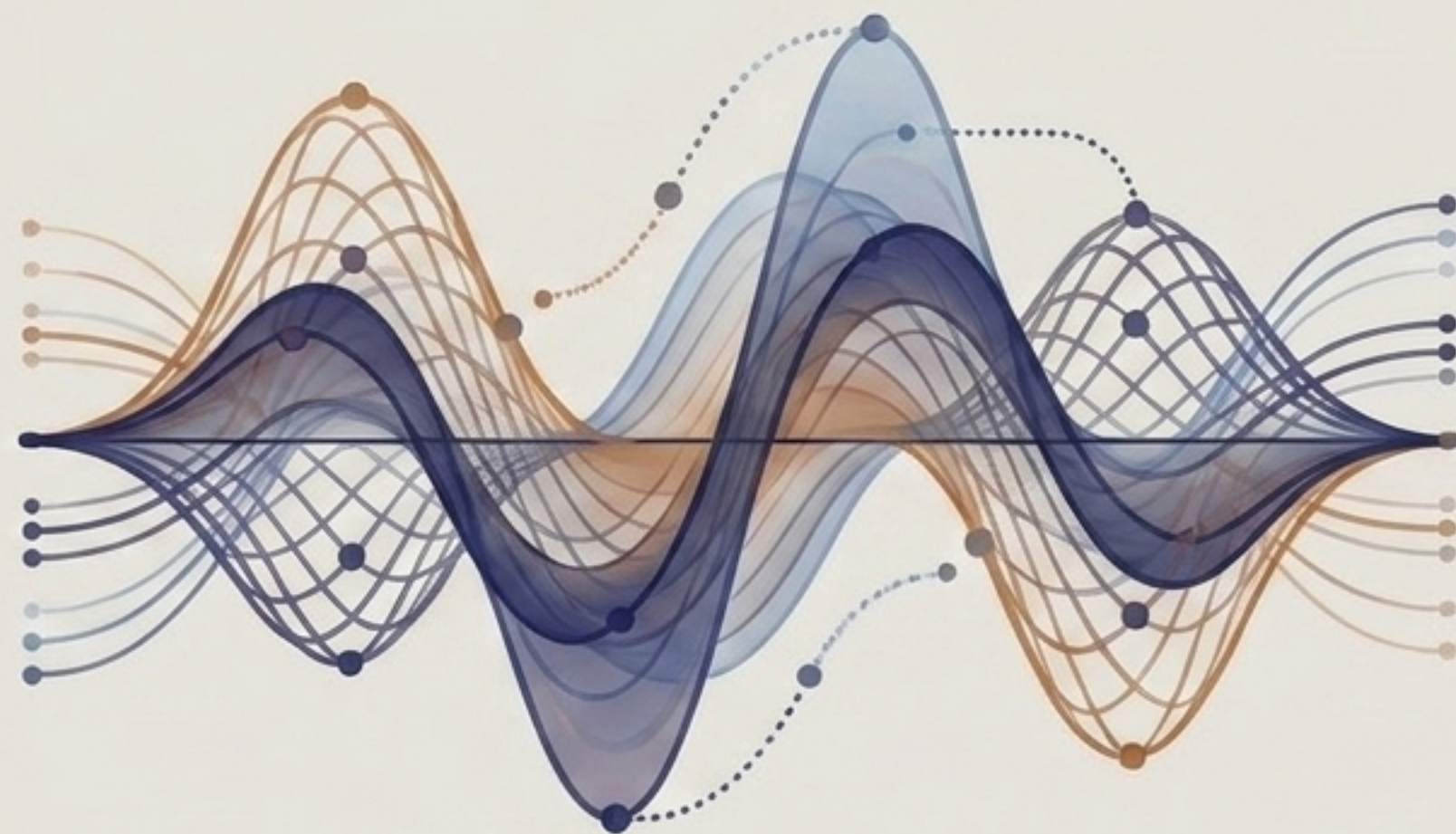
The Shift to Dynamic Trust

Binary Trust



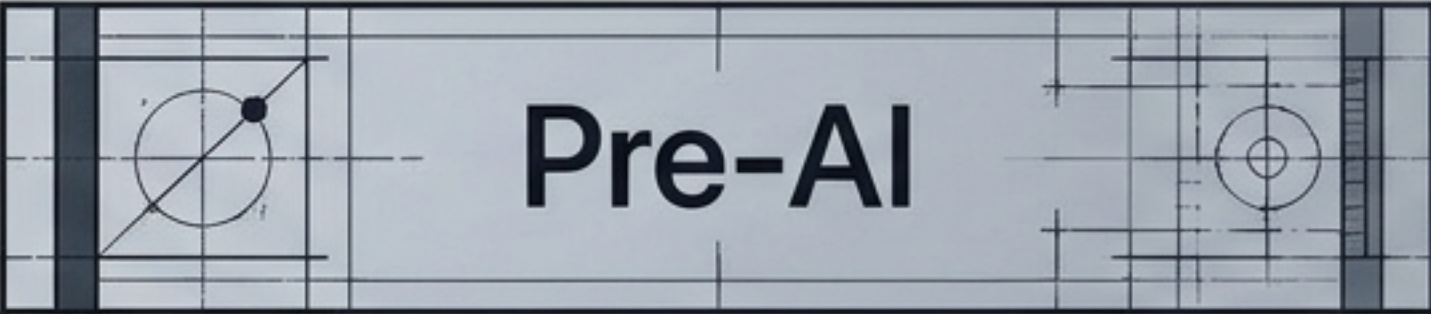
Traditional systems trusted fixed identities: Authenticated users, signed binaries, encrypted channels. A binary Yes/No decision.

Dynamic Trust



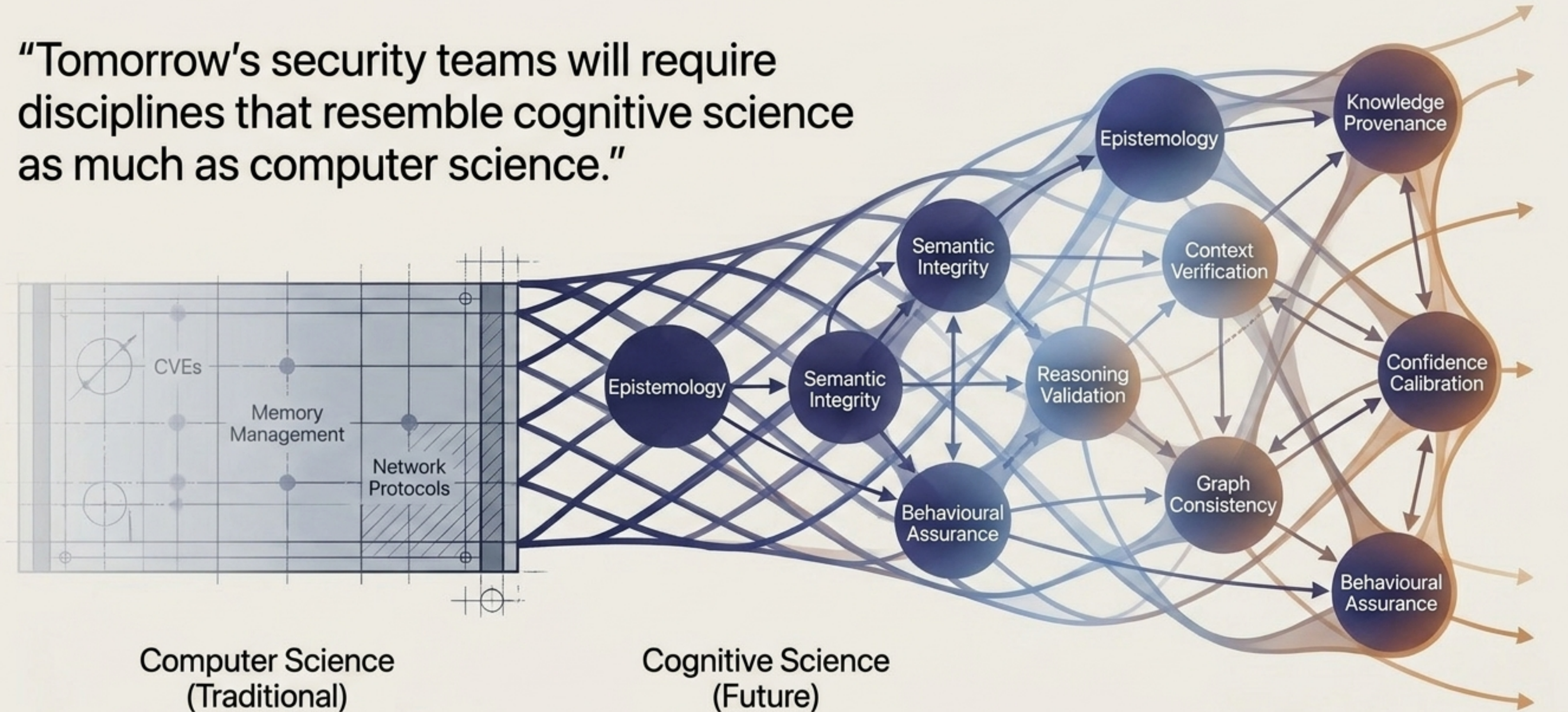
AI systems must continuously evaluate: Information sources, retrieved documents, contextual relevance, inferred intent, and collaborating agents. Trust is an ongoing property.

The Era of Security: Pre-AI vs. Post-AI

	 Pre-AI	 Post-AI
Core Objective	Software Correctness (Can an attacker execute code?)	Behavioural Correctness (Can an attacker influence reasoning?)
What is Trusted	 Verified Code	 Verified Context
Failure State	System Crash / Specification Violation	Safe Operation / Harmful Conclusion
Primary Defense	 Fixing Implementation Flaws	 Verifying Information Integrity

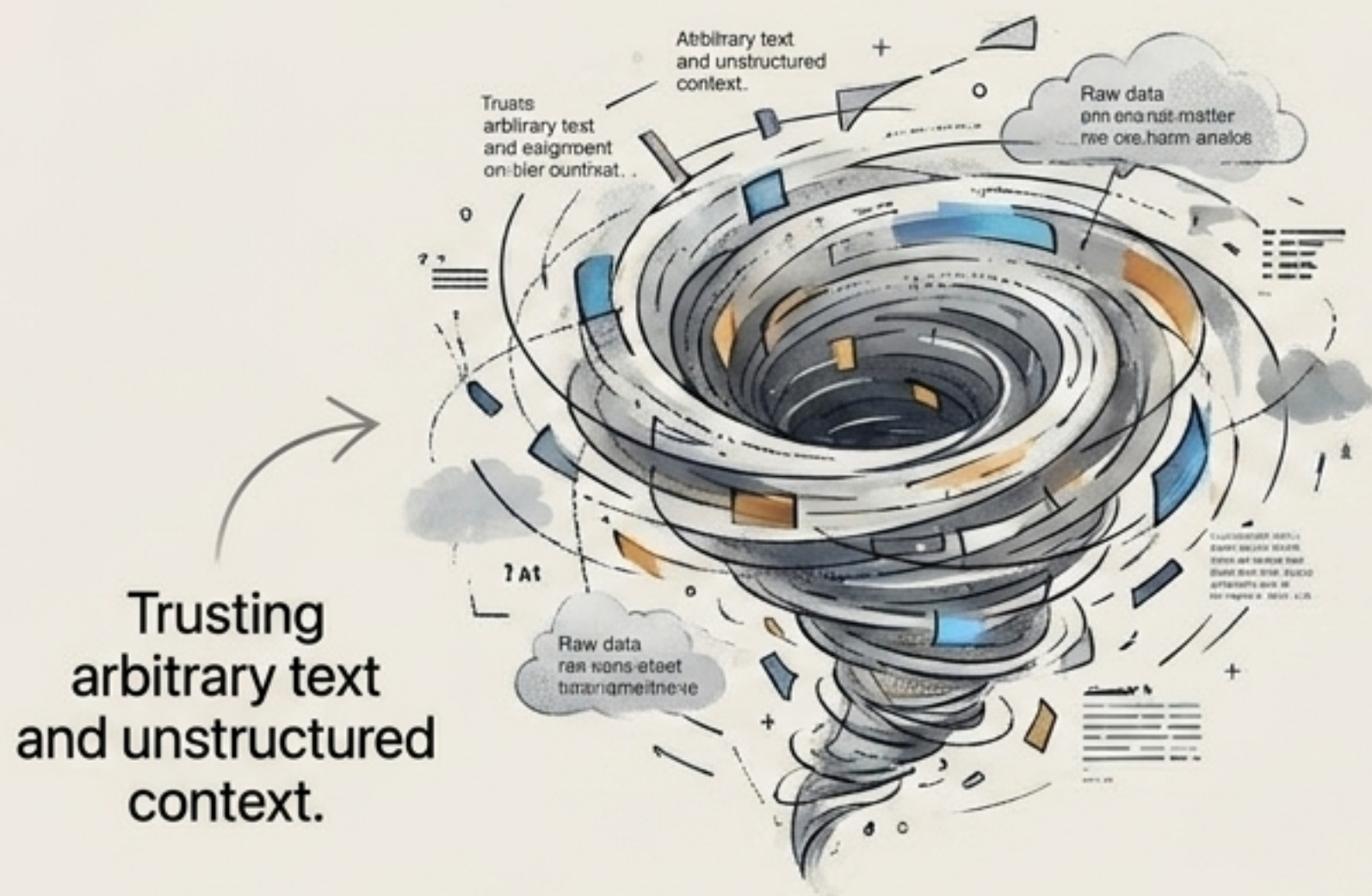
Diagnosing the New Security Disciplines

"Tomorrow's security teams will require disciplines that resemble cognitive science as much as computer science."

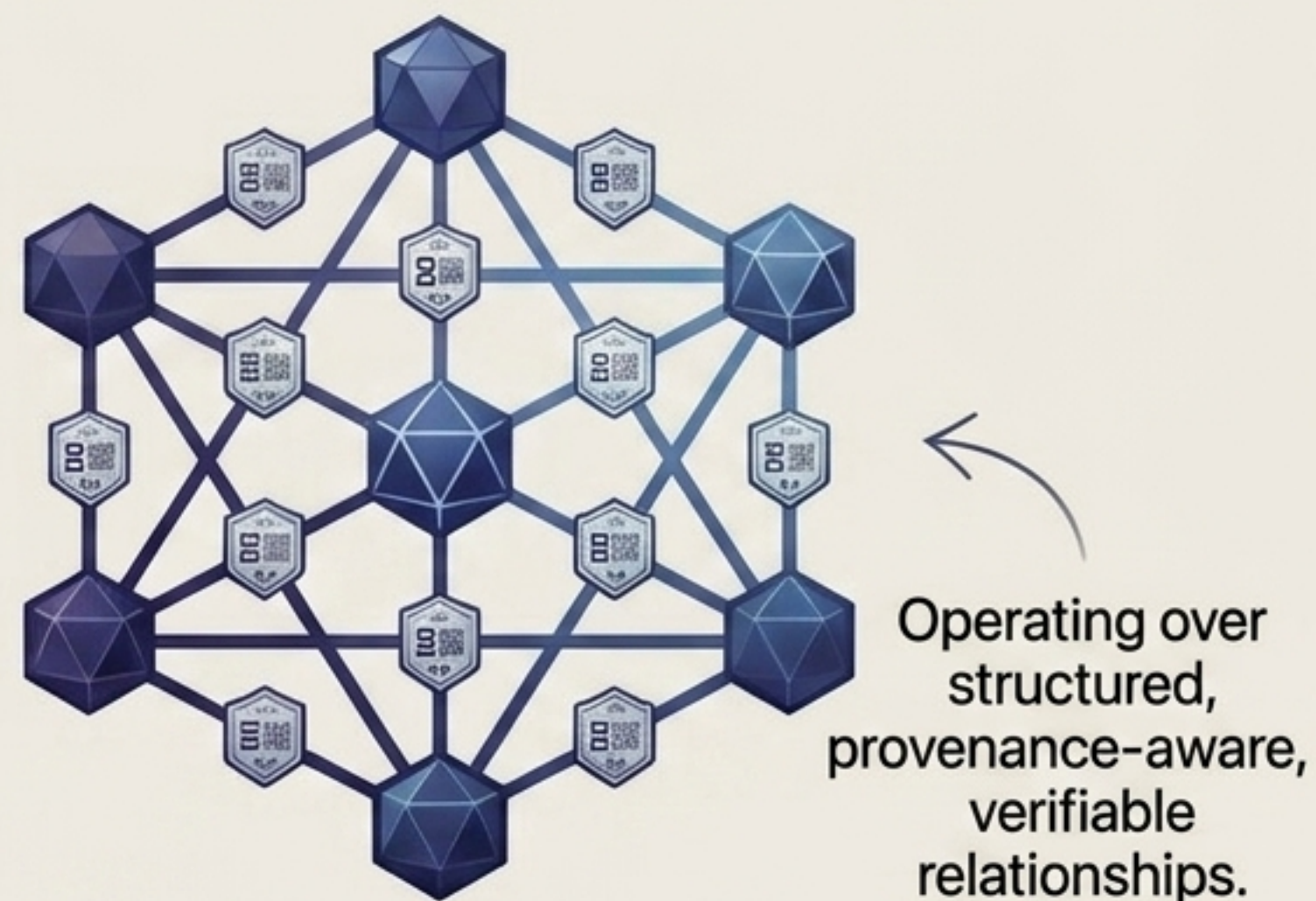


The Solution: Verified Context Graphs (VCGs)

Before (The Vulnerable State)



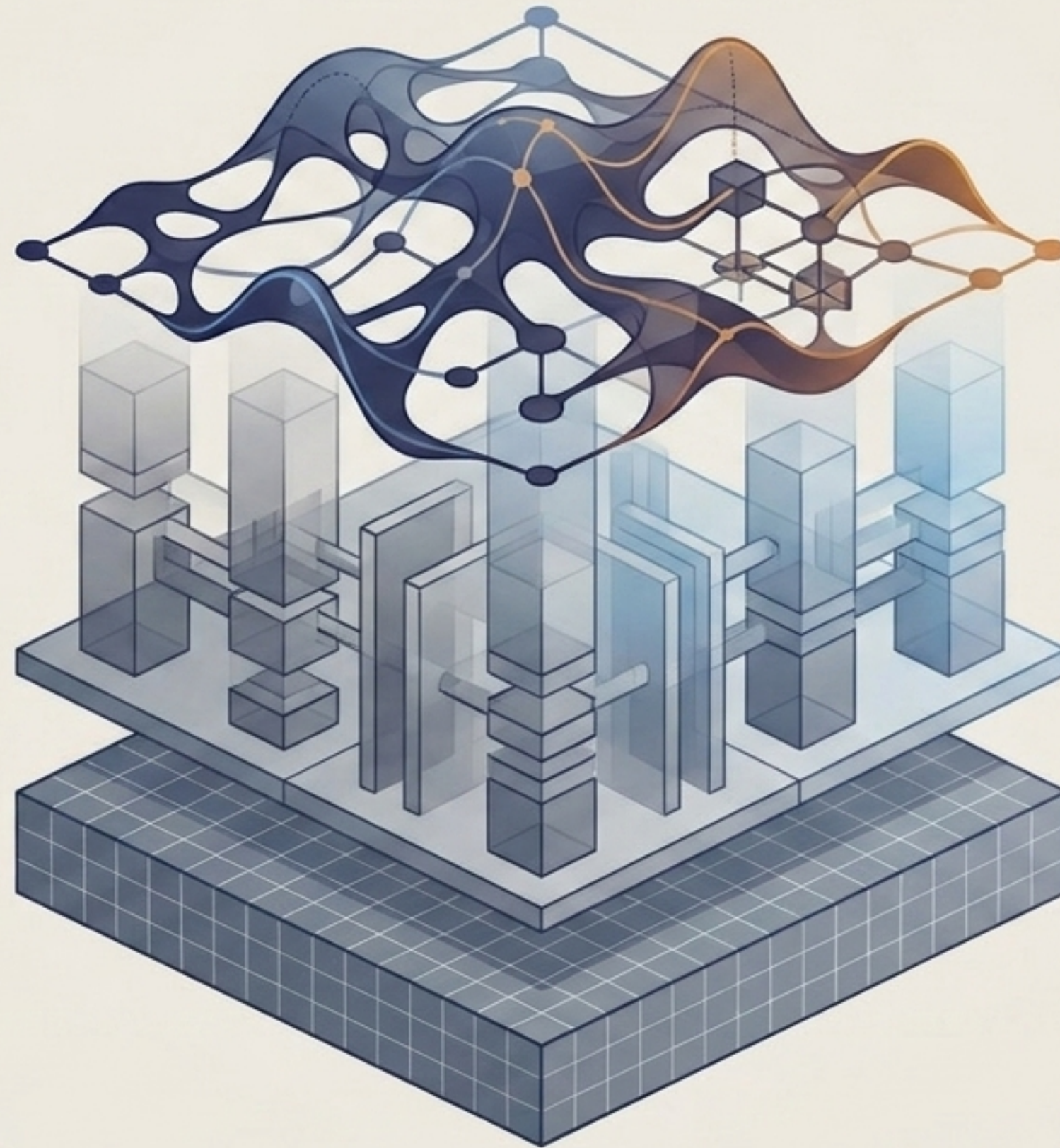
After (The Secured State)



The security boundary moves from software execution to information integrity.
This stops hallucination, prompt manipulation, and context drift.

The Future AI-Native Security Stack

Only together do these layers provide meaningful security in the Age of AI.



The Decision Matrix

Layer 3: Verified Intelligence

Reasoning processes that can explain, constrain, audit, and justify conclusions before acting.

The Pipelines

Layer 2: Verified Information

Trusted, provenance-aware contextual data with explicit relationships (VCGs).

The Foundation

Layer 1: Correct Software

Software free from implementation vulnerabilities.

Security Must Evolve

The defining challenge of the next decade is not only software assurance, but intelligence assurance.

We must evolve from protecting programs to protecting understanding, in an age where intelligence itself is both the platform and the target.